PrP-lowering divalent siRNA 2439-exNA

Information Package for Pre-IND Meeting
Type B Meeting

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1. Executive Summary

This meeting is to discuss an investigator-initiated effort to advance PrP-lowering divalent siRNA 2439-exNA to a first-in-human trial in symptomatic prion disease patients.

Prion disease is a rapidly progressive neurodegenerative disease. From first symptoms, patients steeply decline into profound dementia, losing faculties over the course of weeks and typically dying within 5 months. The disease is universally fatal, and there currently exists no standard of care.

The pathogenesis of prion disease is well understood. A single protein, prion protein (PrP), encoded by the prion protein gene (*PRNP*) misfolds into a protein-only pathogen which spreads across the brain, killing neurons. PrP is strictly required for disease, as evidenced by decades of genetic experiments in animal models. More recently, we have shown with PrP-lowering antisense oligonucleotides (ASOs) that lowering PrP levels in the brain both delays onset and slows progression of prion disease across a range of treatment paradigms.

Both animal and human genetic data indicate that PrP reduction is well tolerated. Homozygous PrP knockout mice have no striking phenotype beyond their well-established invulnerability to prion disease. Evidence suggests a subtle age-dependent peripheral neuropathy is associated with complete loss of PrP but not with 50% reduction. PrP depletion has no known effect on the central nervous system (CNS).

On November 14, 2017, we met with 25 FDA scientists in the context of a Critical Path Innovation Meeting to discuss clinical strategies for the advancement of PrP-lowering therapeutics, motivated by the prospect of development of an ASO therapeutic for prion disease. To the best of our knowledge, an ASO for prion disease remains in active, albeit delayed, development with Ionis Pharmaceuticals. Dr. Vallabh is a patient-scientist living at known high risk for genetic prion disease; we are highly motivated to seek swifter development of PrP-lowering therapeutics on behalf of patients who will not live to see a commercial drug become available.

For the past 3 years, we have collaborated with University of Massachusetts Medical School (UMass) on an NIH-funded project to develop divalent siRNA (di-siRNA) against *PRNP*. Di-siRNA is a fully chemically modified siRNA modality first reported in 2019 that shows promising potency, biodistribution, and durability in the CNS. We have obtained positive survival data in prion-infected mice using a di-siRNA compound directed against mouse *Prnp*, and we have identified lead compound 2439-exNA, directed against human *PRNP*. The potency of this compound at lowering human PrP in transgenic mice indicates, based on everything we know today, high potential to modify disease in human prion disease patients. We are therefore seeking to advance this modality swiftly to patients in need through a small investigator-initiated trial.

We are proposing a small (15 patients) investigator-initiated trial in symptomatic patients diagnosed with prion disease. Because the di-siRNA modality is new to humans, we believe that it should be tested first in symptomatic patients rather than pre-symptomatic at-risk individuals. As discussed, symptomatic patients face dementia within weeks and death within months. Our proposed development plan accounts for the bleak prognosis for these patients, together with the evidence that PrP lowering is an effective strategy and that 2439-exNA shows therapeutic potential.

The goal of the requested meeting is to identify a feasible path forward to a first-in-human trial for 2439-exNA. The specific questions we have posed to the agency are 1) whether our proposed clinical plan is acceptable, 2) whether our proposed toxicology plan, which relies on GLP rat and non-GLP

mouse data only, without exposure in non-human primates, is acceptable, and 3) whether our proposal to differentially size the aliquots of Drug Product used for dosing and for testing is acceptable. Specific questions aside, the meeting will be a success if a path forward is identified.

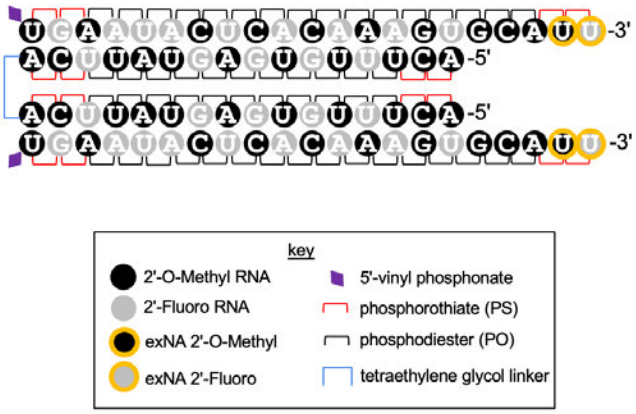
2. Introduction

2.1. Product Name and Application Number

PrP-lowering divalent siRNA 2439-exNA, application number to be assigned.

2.2. Chemical Name and Structure

The product is a divalent siRNA sequence-targeted to human *PRNP*, with a chemical backbone similar to the original divalent siRNA scaffold described by Alterman et al[1], but with a different pattern of 2'-O-Methyl and 2'-Fluoro modifications and phosphoriothioate linkages, and a novel internucleotide linkage called extended nucleic acid (exNA). The composition of 2439-exNA is depicted graphically below.



Alternatively, we define the following nomenclature: 2'-OMe Adenosine, (mA); 2`-OMe Cytidine, (mC); 2`-OMe Guanosine, (mG); 2'-OMe Uridine, (mU); 2'-Fluoro Adenosine, (fA); 2'-Fluoro Cytidine, (fC); 2'-Fluoro Guanosine, (fG); 2'-Fluoro Uridine, (fU); extended 2'-OMe Uridine, (mxU); extended 2'-F Uridine, (fxU) 5'-vinyl phosphate, VP; phosphorothioate linkages, #. With this nomenclature, the antisense strand is 5`-V(mU)#(fG)#(mA)(fA)(fU)(fA)(mC)(fU)(mC)(fA)(mC)(fA)(mA)(fA)(mG)(fU)(mG)(mC)(mA)#(mxU)#(fxU) and the sense strand is 5`-(mA)#(mC)#(mU)(fU)(mU)(fG)(mU)(fG)(mA)(fG)(mU)(mA)(mU)(fU)#(mC)#(mA), with the two sense strands connected by a tetraethylene glycol linker.

2.3. Indication(s)

Prion disease — symptomatic patients.

2.4. Dosage Form, Route of Administration, and Dosing Regimen

50 – 200 mg diluted in artificial CSF as a one-time intrathecal injection.

3. Purpose of the Meeting

The purpose of the meeting is to determine a pathway for submitting an investigator-initiated IND to administer a single dose of this product to a small number of symptomatic prion disease patients. At a high level, we seek to identify a development plan that does justice by the patient population: patients expected to progress to profound dementia within weeks and death within a few months, for whom no standard of care exists and no alternative experimental therapies are currently in trials.

Below, we have noted specific questions to the agency based on potential pathways we have considered, but above all, this meeting will be a success if a feasible path forward is identified.

We also cross-reference PIND 141250 in which we have discussed with CDER the topic of PrP-lowering therapeutics in general, but we note that unlike discussions under 141250, the present meeting does not concern antisense oligonucleotides (ASOs) nor pre-symptomatic at-risk individuals.

4. Background

4.1. Development program to date

Brief introduction to prion disease

Prion disease is a fatal and incurable neurodegenerative disease caused by prion protein (PrP)[2]. PrP is encoded by a chromosomal gene, *PRNP*, and is expressed in the brains of all humans and all mammals. In prion disease, PrP undergoes post-translational misfolding into a self-replicating conformer known as a prion. Prions spread exponentially throughout the brain by recruiting other molecules of PrP, eventually accumulating to a level where they kill neurons. Subtypes of prion disease known historically by various names including Creutzfeldt-Jakob disease (CJD), fatal familial insomnia (FFI), and Gerstmann-Straussler-Scheinker (GSS) disease all share this fundamental molecular mechanism. Prion disease typically presents clinically as a rapidly progressive dementia[3]. 85% of cases are sporadic, apparently due to stochastic protein misfolding events. 15% are genetic, due to gain-of-function mutations in *PRNP*[4], and <1% of cases are acquired through exposure to exogenous prions via dietary or iatrogenic routes[5]. All available data from human genetics, biochemical approaches, laboratory animal models, and observations in wild and agricultural animals that are natural hosts of prion disease, are unanimous that PrP is the cause of prion disease. The therapeutic hypothesis of PrP lowering is backed by decades of genetic experiments in animal models and more recently by preclinical proof-of-concept studies in mice using antisense oligonucleotides (ASOs)[6–14] (see below), but to date, no therapy exists, and no PrP-lowering drug candidate has ever entered human trials.

We have previously interacted with FDA under PIND 141250 about PrP-lowering ASOs, and in particular about the potential for a pathway to treat pre-symptomatic individuals at risk for prion disease. The present submission, however, focuses on already-symptomatic patients. Prion disease is uniformly fatal, and for the most common subtype, sporadic CJD, median survival is 5 months from first symptom[15]. During this time, cognitive and functional decline is extremely rapid: most patients

lose the ability to complete all basic activities of living such as speaking, walking, eating, and using the toilet, within weeks[16]. Moreover, most patients lose considerable time just seeking a diagnosis — historically prion-specific diagnostic tests were not even ordered until ~3 months into the disease course[17] — though there is some evidence that diagnosis is improving in recent years[18]. In previous randomized trials, patients died a median of ~2 months from randomization[19,20], and many were profoundly impaired well before this time[21]. Our own experience of prion disease both personally and through the patient community impels us to be bold about advancing a rational therapeutic into first-in-human studies.

The foundation for PrP lowering

PrP causes prion disease. Decades of work from multiple orthogonal approaches including human genetics, histological and biochemical approaches, and animal models, all converge on PrP as *the* molecular cause of prion disease (Figure 1). Briefly: infectious prions are composed of misfolded PrP[2,22,23]; PrP deposition is pathognomonic for prion disease neuropathology[24]; prion strains are encoded in PrP conformation[25–27]; PrP genetic variants cause genetic prion disease, impact sporadic prion disease risk, affect prion disease duration and presentation, govern species barriers and affect transmission risk in humans and other animals[28]. Because all prions, regardless of strain or species, are comprised of PrP, it has long been expected that depleting PrP substrate should be a universal therapeutic for prion disease regardless of the patients' etiology (sporadic, genetic, acquired) or subtype.

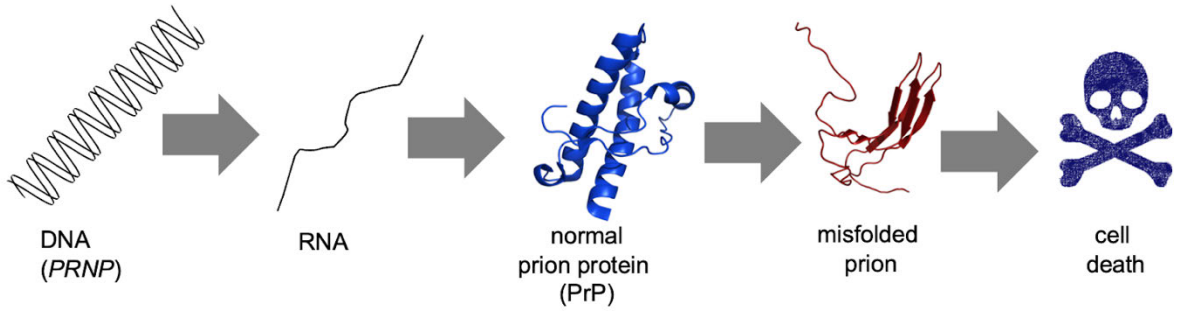


Figure 1. Pathway of prion disease.

PrP lowering is an exceptionally deeply validated therapeutic hypothesis. The advent of transgenic and knockout mice 30 years ago provided the first genetic evidence that PrP dosage governs the tempo of prion disease[7] (Figure 2A). These models are relevant because prion disease naturally afflicts not only humans but also diverse mammals including deer, elk, sheep, goats, cattle, camels, and mink. It is exceptionally well modeled in mice, where intracerebral inoculation of prions into naive wild-type animals leads to fatal disease within 5-6 months, with all of the same pathological hallmarks seen in the human disease[29]. Heterozygous knockout mice survive prion infection more than twice as long as wild-type mice (Figure 2A), and homozygous knockouts are incapable of propagating prions or developing prion disease[6,13]. Conversely, disease is accelerated in transgenic mice overexpressing PrP[7]. This finding has replicated in conditional genetic models, where lowering or abolishing PrP expression can respectively delay or prevent development of disease[9,10]. More recently, we showed that chronic administration of an ASO lowering PrP by ~50% can triple survival in prion-infected mice[12] (Figure 2B). This result was obtained as long as therapy is initiated at any time from prior to infection up through 78 days post-infection (dpi), when animals are presymptomatic but neuropathological sequelae including astrogliosis and increase in plasma neurofilament (NfL) are emerging[12]. Survival benefit in early-treated animals was comprised mostly

of an extension of healthy life. In animals treated at the cusp of frank symptom onset, a single dose of ASO was still able to increase survival by 70% (Figure 2C)[12]. Benefit is directly responsive to PrP dosage: a low dose of ASO corresponding to 21% knockdown of *Prnp* RNA in cortex led to a significant 15% increase in survival time, but higher doses achieving deeper target engagement increased survival by larger amounts[12]. Pharmacologic reduction of PrP in the brain was reflected in the CSF in rats, indicating that CSF PrP quantified by ELISA could serve as a pharmacodynamic biomarker[30].

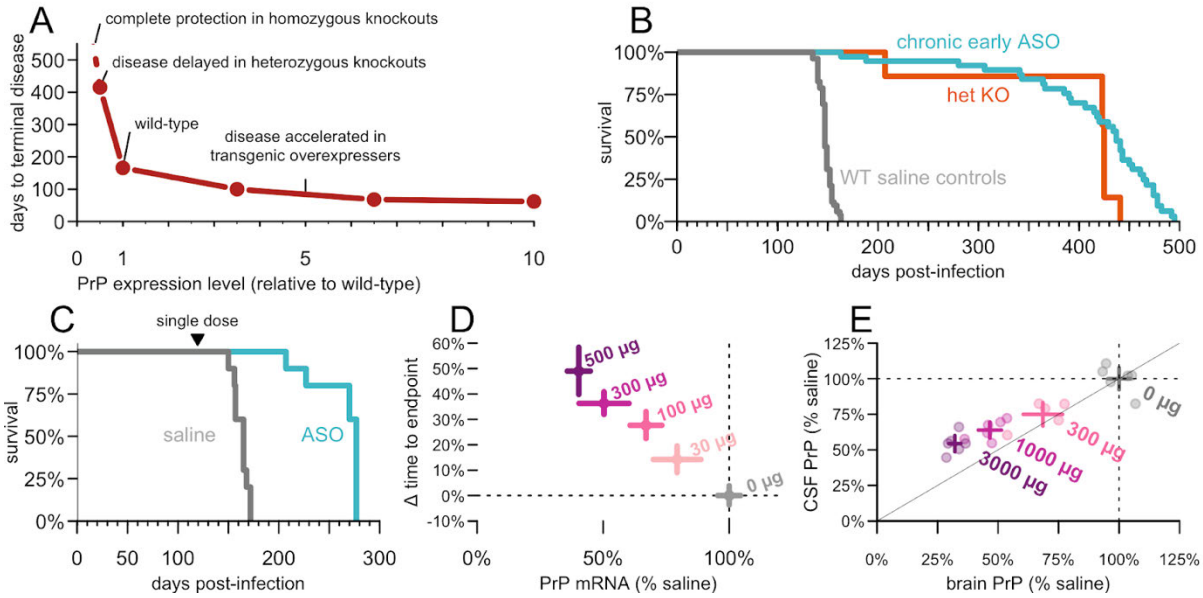


Figure 2. Published genetic and pharmacologic proofs-of-concept validating PrP lowering as a therapeutic hypothesis in prion disease. A) *Effect of PrP expression level on survival time after intracerebral prion inoculation in mice across wild-type, transgenic, and knockout lines.* ***B)*** *Benefit of chronic early ASO treatment on a symptomatic endpoint mirrors heterozygous knockout in mice.* ***C)*** *A single ASO dose near the onset of symptoms delays terminal endpoint in mice.* ***D)*** *Continuous dose-response relationship between PrP and time to symptomatic endpoint in mice.* ***E)*** *CSF PrP mirrors brain PrP knockdown after ASO administration in rats. Panel A adapted from ref.* [14]*, B-D from ref.*[12]*, E from ref.* [30]*.*

We felt it prudent to also validate the efficacy of PrP lowering across strains and across species of PrP. We have already published that PrP lowering by either heterozygous genetic knockout or by ASO administration was efficacious against five murine prion strains of diverse origins — human, agricultural (sheep/goat scrapie), and synthetic[12]. More recently, we created an allelic series of transgenic mice expressing human PrP at different levels: Tg26372 (10 and 20 HuPrP transgene copies in heterozygotes and homozygotes) and Tg25109 (3 HuPrP transgene copies) (Figure 3A). After screening several isolates (Figure 3B) we identified an isolate of sporadic CJD type MM1, the most common subtype of prion disease, that transmitted readily to these mice, causing fatal disease (Figure 3B) and reaching terminal prion titers of 10^11 infectious units per gram in the brain (Figure 3C-D). Time to develop disease was inversely related to PrP protein expression level in these animals (Figure 3E), and animals with lower expression levels maintained weight (Figure 3F) and nest-building activity (Figure 3G) for longer. These observations validate that PrP lowering is relevant to human prions. We also note that the resistance of PrP knockout animals to prion disease has been confirmed for sheep and cattle as well[31,32].

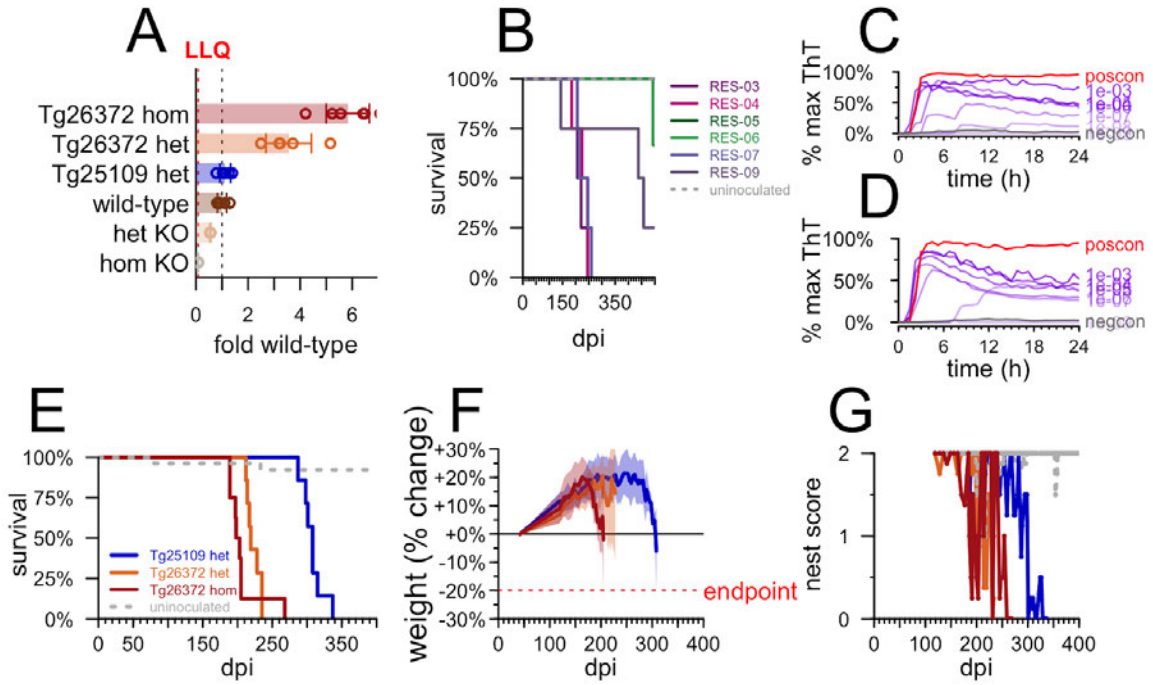


Figure 3. Validation of therapeutic hypothesis in humanized mouse models. A) *Expression levels in cohorts of 5-6 transgenic animals per genotype compared to WT controls (het KO and hom KO controls included to demonstrate assay selectivity).* ***B)*** *Challenge of groups of N=4 Tg26372 homozygous animals with human prion isolates.* ***C-D)*** *RT-QuIC kinetic curves (standard conditions) for a RES-03 inoculated animal (C) and a RES-07 inoculated animal (D).* ***E)*** *Survival of cohorts of 7-8 animals infected with RES-03 as a function of genotype and thus PrP expression level. A total of 26 uninoculated animals across the 3 genotypes were used as controls; monitored in the same experiment were F) weight change from individual baseline and G) nest-building, scored as reported*[12]*.*

In addition to PrP's indisputable central role in prion disease, we are fortunate that PrP appears dispensable for healthy life. PrP knockout mice[33], cows[31], and goats[34] are all healthy and viable and initially defied efforts to identify any knockout phenotype other than resistance to prion infection. It was eventually discovered that homozygous (but not heterozygous) PrP KO mice exhibit a myelin maintenance deficiency in peripheral nerves (but not in the CNS) leading to a mild peripheral neuropathy[35]. This is apparently due to a lack of signaling of an N-terminal PrP peptide to Adgrg6, a receptor expressed on peripheral Schwann cells but not in the CNS[36]. While this myelin phenotype is histologically evident in aged animals, it is phenotypically subtle. Identifying a difference in grip strength, rotarod performance, or hot plate response is difficult, and some aged (60 week) PrP KO mice but not others exhibit a small but significant difference from wild-type animals, apparently as a function of mouse strain[35]. The histological phenotype has further replicated in PrP KO goats, where no outward phenotypic difference was reported[37]. We have identified older humans with naturally occurring heterozygous loss-of-function mutations in *PRNP* who appear to be healthy[4]. Moreover, such mutations occur in the population at exactly the frequency expected based on mutation rates; unlike the vast majority of genes in the human genome, there appears to be no purifying selection against loss-of-function variants in *PRNP*[38]. While Phase I studies should be conducted with the usual high level of care and monitoring, we have no reason to expect on-target toxicity from lowering PrP to the minimum achievable level.

Development of PrP-lowering divalent siRNA

Divalent siRNA (di-siRNA) was first described in 2019 as a novel oligonucleotide modality with desirable potency and durability properties for lowering a target RNA in the CNS[1]. In early 2020, our team at the Broad Institute partnered with Dr. Anastasia Khvorova's team at UMass Medical School to screen and develop divalent siRNAs for prion disease. We applied for and were awarded an NIH grant (NINDS IGNITE R61/R33 NS119717) to support screening and preclinical proof-of-concept studies. UMass licensed its divalent siRNA patent portfolio to a biotechnology startup, Atalanta Therapeutics, which also collaborated and advised on our NIH-funded project. As promising data (see below) accumulated, we hoped that Atalanta would develop a di-siRNA for prion disease as a mainstream commercial program. Unfortunately, in August 2022 the company informed us that, for business reasons, they could not commit to carrying the prion program forward. Instead, they offered us a letter pledging that they would not use their IP to block our own development of a therapeutic. We therefore began to explore our options to leverage federal and/or philanthropic funding sources to bring di-siRNA forward as an investigator-initiated project, and in October 2022 applied for another NIH grant (U01 FOA PAR-22-030). Our proposal received a favorable Impact Score of 13, and we are awaiting a funding decision from NIH. If funded, this grant will provide some, albeit limited, support for IND-enabling studies.

While we continue to hold out hope that a PrP-lowering ASO may enter clinical trials, and indeed could ultimately prove effective, we believe that is crucial to bring di-siRNA forward as an additional drug candidate in the pipeline. In our proof-of-concept mouse studies, we were able to lower PrP only ~50% with ASOs, an amount sufficient to delay but not prevent or reverse prion disease; all animals eventually succumbed to terminal prion disease. In addition, significant questions about toxicology and brain distribution remain for ASOs. The ASO program about which we previously interacted with FDA is now years behind schedule, and data from CNS ASO trials over the past few years in other indications have not always been promising. Given that the mouse data for PrP-lowering di-siRNA look very encouraging, we feel we have a mandate to find a way to advance this therapeutic in parallel, and not merely wait for PrP-lowering ASOs to progress.

Proof-of-concept data

Divalent siRNA background and chemical scaffold. Divalent siRNA (di-siRNA), invented by co-investigator Anastasia Khvorova, consists of two fully chemically modified siRNA duplexes connected by a tetraethylene glycol linker[1]. This modality was tested in mice and in non-human primates and provided exceptionally deep lowering of model target huntingtin (*HTT*), with sustained activity out to 6 months[1]. Dr. Khvorova's lab has discovered a new (as yet unpublished) nucleic acid chemical modification called extended nucleic acid (exNA), in which the phosphodiester (PO) or phosphorothioate (PS) backbone is extended by one methyl group at the 5' (Figure 4A). The superior nuclease resistance of this modification (Figure 4B) inspired the design of a new di-siRNA scaffold with reduced PS content, hypothesized to offer excellent potency and a wide therapeutic index (Figure 4C). The exNA siRNAs described herein also incorporate two further improvements over first-generation di-siRNAs. First, these ExNA siRNAs incorporate a "fixed tail" where the two exNA bases are converted to a U rather than matching the target sequence. The terminal fU enables best fit in the PAZ domain of the RISC complex and lack of direct complementarity to the target disables the mechanism of loaded RISC degradation, thus enhancing potency and duration of effect. Second, the pattern of 2'OMe and 2'F modifications has also been optimized[39] compared to the first-generation di-siRNAs. The final chemical scaffold is shown in Figure 4C.

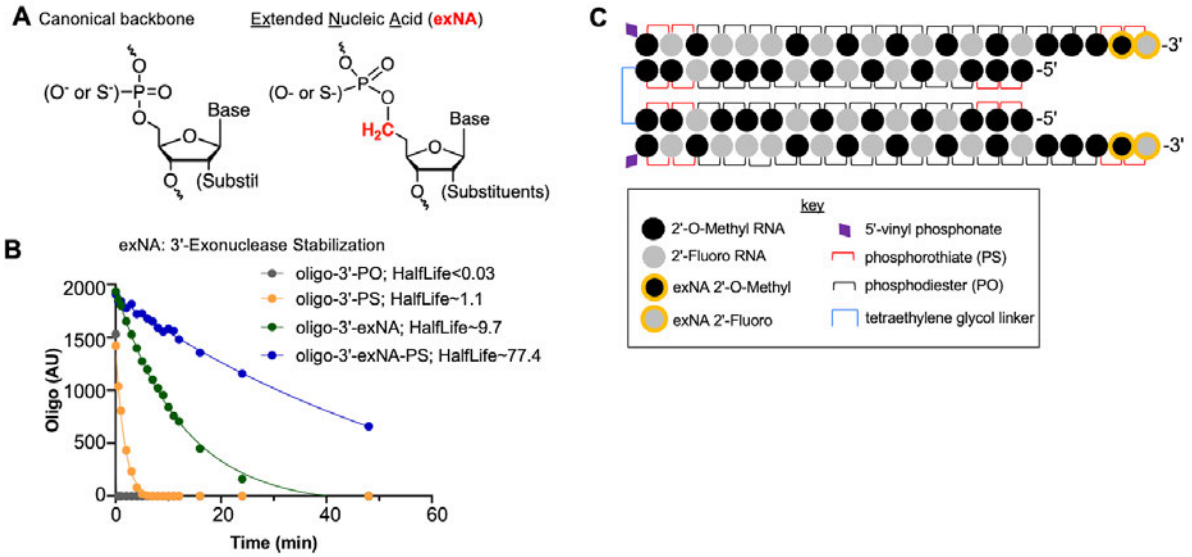


Figure 4. Extended nucleic acid divalent siRNA. A) *Identity of the exNA modification. B) Increased exonuclease resistance of exNA-modified oligonucleotides. C) Divalent siRNA chemical scaffold with exNA and reduced phosphorothioate content.*

Proof of concept studies with mouse tool compounds. To identify tool compounds for initial studies, we performed a small-scale screen in mouse N2a cells of 20 siRNA sequences targeting mouse *Prnp* (Figure 5A). This identified oligo 1682 as promising, with 11.7% residual *Prnp* mRNA. We synthesized this molecule in divalent format in the new exNA scaffold (Figure 4C), and compared its potency versus saline or a non-targeting control compound after a single ICV dose in wild-type C57BL/6N mice (Figure 5B). For clarity, note that all mouse doses described throughout this document are in moles of divalent siRNA compound (1 mol di-siRNA = 2 mol monovalent siRNA equivalents) and are given as total doses per animal (10 nmol total dose is administered via bilateral ICV, so 5 nmol per side). At 30 days post-dose 1682-exNA at 10 nmol lowered the target by half, leaving 49.4% residual PrP protein in whole brain hemispheres measured by ELISA[30]. This is similar to the potency of PrP-lowering ASO tool compounds we have tested previously. (Note that the human di-siRNA lead now identified for advancement is more potent; see Figure 6.) We advanced this compound into studies in a standard disease model: wild-type mice infected intracerebrally with the RML strain of mouse prions.

Divalent siRNA treatment was initiated at different timepoints in two cohorts: 75 days post-inoculation (dpi), corresponding to a pathological disease stage (evinced by elevated neurofilament and astrogliosis), and at 126 dpi, approximately at the onset of symptoms. 75 dpi animals received chronic treatment every 120 days (q120d) while 126 dpi animals received a single dose. At each dosing timepoint, animals received 10 nmol of divalent siRNA (20 nmol monovalent equivalents) split between bilateral doses into CSF via the cerebral ventricle. Control animals received a non-targeting control exNA compound, saline, or no injection.

In the 75 dpi treatment study (Figure 3A),1682-exNA animals lived 2.7 times longer than saline controls (median 442 vs. 165 dpi, P = 0.001, log-rank test), eventually developing typical prion disease signs and succumbing to weight loss endpoint. That this outcome is similar to that achieved with ASO tool compounds of similar potency, despite less frequent dosing (q120d here vs. q90d in the published ASO study) may point to superior durability of di-siRNA. In the 126 dpi single dose cohort, 1682-exNA animals lived 1.6 times longer than controls (median 270 vs. 164 dpi, P = 0.0002, log-rank test, Figure 3C) before succumbing to weight loss (Figure 3D) — again, survival time was

similar to that of animals that received a single dose of ASO in an analogous published experiment (275±3 dpi)[12]. When considered relative to *remaining* survival time from the moment of treatment at 126 dpi, late treatment increased survival by 3.8-fold. In neither cohort of di-siRNA treated animals did we observe premature deaths after dosing, so the treatment appears well tolerated. These results support di-siRNA as a promising modality for treating prion disease. The benefit of 50% reduction of PrP appears generally similar whether achieved by di-siRNA, ASO, or heterozygous genetic knockout, consistent with our mechanistic understanding of prion disease as strictly dependent upon PrP expression. This confirmed our expectation that for a divalent siRNA to achieve superior therapeutic benefit or superior probability of clinical success in prion disease, compared to an ASO, would depend upon identification of a compound with superior potency, durability, and/or tolerability.

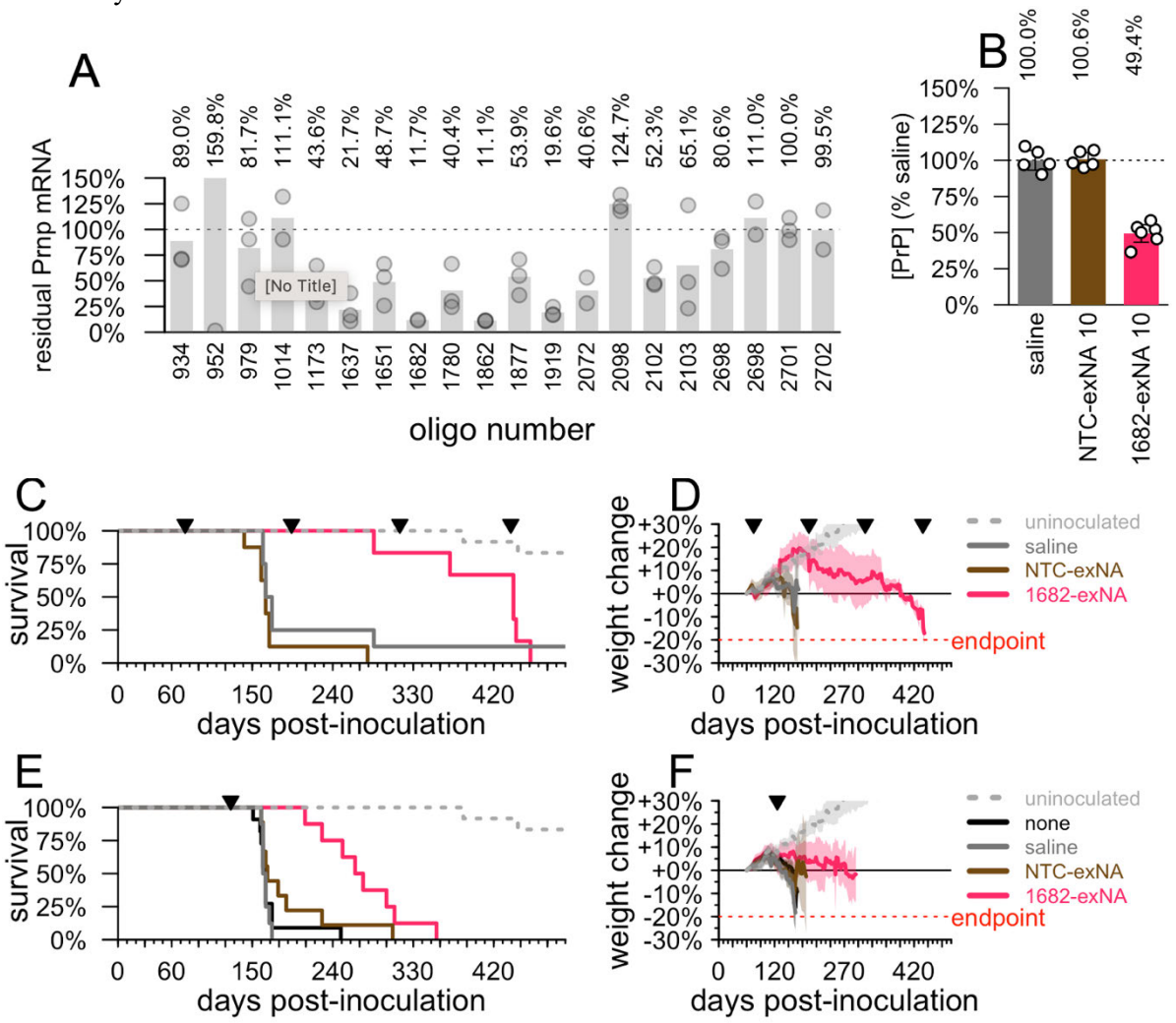


Figure 5. Efficacy of a divalent siRNA tool compound in mice with prion disease. A) *In vitro screen for potent siRNA sequences against mouse Prnp. Screened in triplicate in N2a cells with Prnp mRNA quantified by branched DNA (bDNA) assay, normalized to the mean of untreated cells.* ***B)*** *In vivo test at 10 nmol (total dose, bilateral ICV) in C57BL/6N mice, whole brain hemisphere PrP by ELISA[30], 30 days post-dose.* ***C)*** *All-cause survival of RML prion-inoculated animals treated with 1682-exNA every 120 days beginning at 75 dpi, triangles indicate doses. N=8 per group.* ***D)*** *Weight change trajectories versus individual baseline in animals from (C), lines indicate means and shaded areas indicate 95%CIs.* ***E)*** *All-cause survival of RML prion-inoculated animals treated with a single dose of 1682-*

exNA at 126 dpi, triangle indicates dose. N=8-11 per group. ***F)*** *Weight change trajectories versus individual baseline in animals from (E), lines indicate means and shaded areas indicate 95%CIs.*

Identification of a potent human di-siRNA lead. Because our mouse tool compound, above, exhibited similar potency and similar efficacy as a previously reported ASO, we launched a more intensive screen for candidate human *PRNP*-targeting sequences in order to identify candidates with superior potency. We screened 85 human *PRNP*-targeting siRNAs in human U251-MG glioblastoma cells. At 2 µM test compound, all tested compounds had at least some potency (>10% knockdown or <90% residual target RNA) compared to untreated cells (Figure 6A), with 34 human compounds resulting in >90% knockdown (<10% residual; Figure 1A), a challenge for discrimination of most potent compounds. At 0.5 µM test compound, several human compounds differentiated themselves as most potent, with ten compounds achieving <8% residual (Figure 6B, highlighted in red). 10 hits from the 0.5 µM condition were advanced into in cellulo dose-response to determine IC_{50} values (Figure 2), and ultimately 7 sequences were advanced to in vivo potency screening (the most potent compounds plus others prioritized for reasons such as species cross-reactivity).

For in vivo potency screening, we tested compounds in either the novel exNA scaffold (Figure 3C) or in the originally reported high phosphorothioate (hiPS) scaffold or a variant thereof (P3V1), along with saline and non-targeting control (NTC) di-siRNAs of each scaffold. Compounds were injected at either 1 or 10 nmol in Tg26372 homozygous mice (Figure 3A) which possess 20 copies of human *PRNP* and express ~6-fold human PrP levels, on a background of endogenous PrP knockout. After 30-34 days in-life, animals were sacrificed and whole brain hemispheres homogenized for analysis by our in-house PrP ELISA[30]. Compound 2439-exNA emerged as the most potent, with 28.3% residual PrP. Notably, for all 6/6 sequences where both scaffolds were tested, the exNA scaffold provided superior knockdown to the hiPS/P3V1 scaffolds.

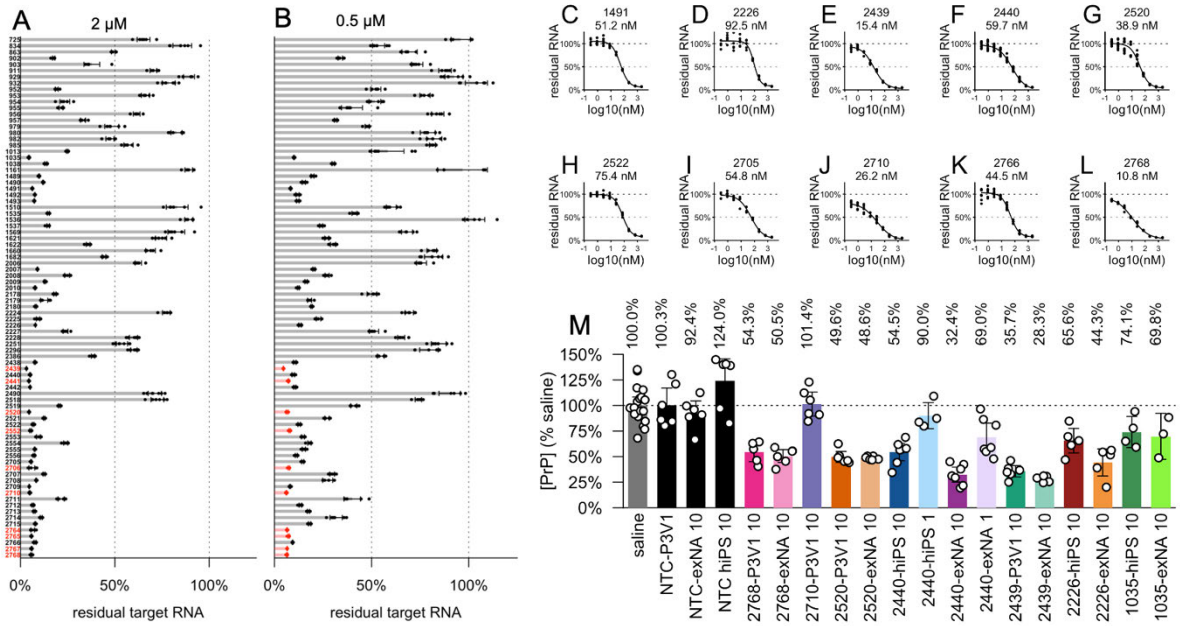


Figure 6. Identification of human di-siRNA lead sequence 2439. M) *All human compounds screened to date, each in groups of 3-7 Tg26372 humanized mice, whole brain hemisphere PrP quantified by ELISA and normalized to within-cohort, within-plate saline controls. NTC=non-*

targeting control. hiPS/P3V1 = high phosphorothioate backbones, exNA = low phosphorothioate + exNA backbone. Numbers after compound indicate total nmol per animal of divalent compound.

To confirm whether the exNA scaffold was indeed superior, we designed a head-to-head in vivo study using lead sequence 2439. This experiment had the goals of i) confirming whether exNA siRNAs achieve superior knockdown, ii) determining which brain regions drive superiority, and iii) determining whether the exNA nucleotides per se, or the UU "fixed tail", drives the superiority. Groups of N=8 Tg26372 humanized mice per cohort received any of 11 treatments: saline, or sequence 2439 in the form of 10 nmol fixed tail loPS, or 0.4, 2, or 10 nmol of P3V1 matched tail, P3V1 fixed tail, or exNA fixed tail (Figure 7A). As predicted, the 10 nmol dose of the exNA scaffold proved superior on the metric of whole-hemisphere protein level reduction (Figure 7B). We further analyzed target engagement in 6 brain regions by qPCR (Figure 7C-H). Across 6/6 regions analyzed, 10 nmol exNA scaffold provided the deepest knockdown. In a combined nested ANOVA model controlling for dose, brain region, and individual animal effects, fixed-tail hiPS compounds outperformed matched-tail hiPS by 6.8 percentage points of knockdown (P=0.0005), while exNA compounds were yet superior to fixed-tail hiPS compounds by an additional 7.7 percentage points (P=6e-15). None of the di-siRNA compounds tested resulted in a change in body weight trajectory over four weeks (Figure 7I), and none resulted in a significant increase in neuroinflammatory markers *Gfap* or *Iba1* (Figure 7J, 7K). These results confirm 2439-exNA as the lead human compound.

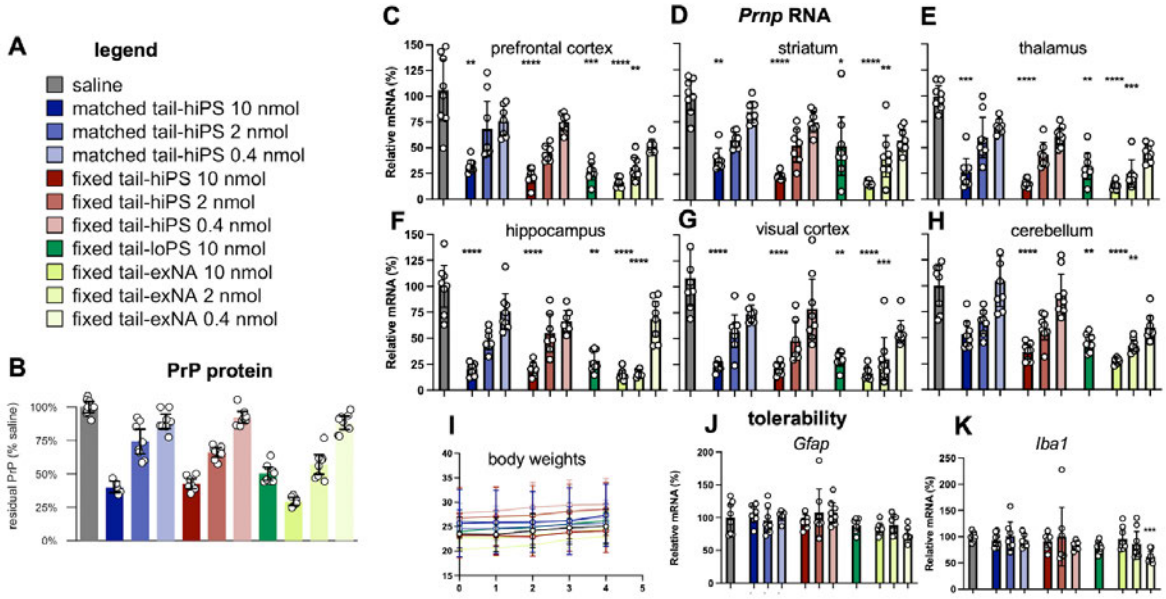


Figure 7. Confirmation of lead scaffold fixed tail-exNA. *Groups of N=8 Tg26372 humanized mice per cohort received any of 11 treatments (legend in **A**): saline, 10 nmol fixed tail loPS 2439, or 0.4, 2, or 10 nmol of 2439 with hiPS matched tail, hiPS fixed tail, or exNA fixed tail.* ***B)*** *PrP ELISA (whole hemispheres).* ***C-H)*** *qPCR for PRNP RNA, 6 brain regions, normalized to Tbp and further to the mean of saline controls.* ***I)*** *Body weight trajectories for individual animals over four weeks post-dose.* ***J)*** *Gfap and* ***I)*** *Iba1 qPCR.*

The target of bases 2-17 of sequence 2439 is perfectly matched in cynomolgus and rhesus macaques (Figure 8). New world monkeys have a single mismatch at base 2. Other potential preclinical species are not matched.

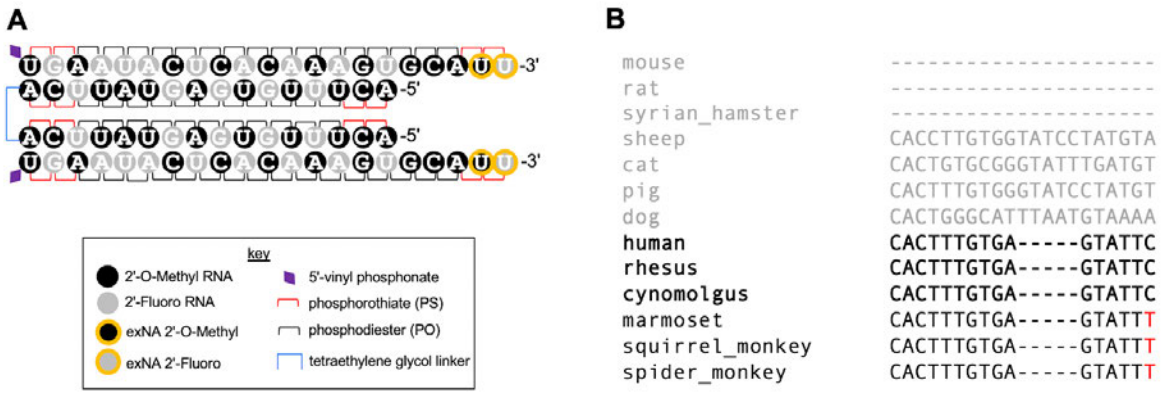


Figure 8. Identity and cross-species reactivity of compound 2439. *A) Structure of lead compound 2439-exNA. B) Alignment of genomic target of bases 2-17 in preclinically relevant species. Red=mismatch.*

Studies underway or planned, funded by our existing NINDS IGNITE grant, include testing of duration of action of 2439-exNA by assessing PrP knockdown at timepoints out to 6 months post-dose, and assessing efficacy of 2439-exNA in prion-infected mice. For biosafety reasons, we have chosen not to test 2439-exNA in humanized mice infected with human prions; instead, we are characterizing a new mouse model with human non-coding *PRNP* sequence but mouse coding sequence, so that we can test human sequence-targeted drug candidates against less biohazardous mouse prion strains.

4.2. Status of product development

2439-exNA is being developed as an academic, investigator-initiated project with NIH and philanthropic funding. There are no plans to commercialize.

4.3. General Investigational Plan

Our approach is non-allele-specific and in principle, our lead drug candidate is capable of lowering PrP in any and all patients diagnosed with or at risk for prion disease. The potential patient population for this approach is therefore hundreds of patients in the U.S. and low thousands worldwide. Pre-symptomatic patients will ultimately require chronic dosing with a PrP-lowering drug in order to prevent disease onset. If a drug is effective, then it will keep symptomatic patients alive long enough that they, too, will require repeat dosing. Therefore, one could contemplate a more conventional development path in which a di-siRNA drug undergoes 9-month chronic dosing in a large cohort of non-human primates and multiple ascending dose trials in multiple patient subpopulations. However, we deemed such an approach infeasible for an academic-led effort. Instead, we believe the unique features of di-siRNA technology together with learnings from ASO development allow us to design a minimalist first-in-human strategy in a small number of sick patients. Our rationale:

1. **The novelty of di-siRNA befits starting with small trials in sick patients.** True, we are vocal proponents of early intervention in healthy, at-risk people[14,40], and we have worked to establish a pathway to primary prevention. However, divalent siRNA has never been tested in humans before, and indeed, no oligonucleotide drug containing exNA linkages has ever been tested in humans before. Thus, the novelty of our drug carries more risk than an ASO and

may not be ideally suited to first testing in healthy people. Instead, we believe there is a mandate to begin in sick patients dying of prion disease before trialing di-siRNA in at-risk people. Moreover, we believe that a small initial trial is appropriate before exposing a larger number of people to the novel drug.

2. **The potency and durability of di-siRNA support a single dose study.** Even our mouse tool compound 1682-exNA, which is of poorer potency than our human lead 2439-exNA, when administered at the onset of symptoms approximately quadrupled the median time that animals had left, from 38 days to 144 days (~4.7 months). In previous randomized trials in prion disease, the median patient died ~2 months from randomization[19,20]. Therefore, in a symptomatic patient population, a single dose of di-siRNA should be sufficient to evince benefit. We propose that beginning with a single dose study should lower the complexity and duration of nonclinical studies required, thus allowing us to gain clinical learnings sooner rather than later. If a single dose is beneficial, we can determine the optimal dosing regimen in subsequent trials, whereas if a single dose is not beneficial, it is unlikely that repeat dosing would be either. We note that ASOs for Huntington's disease and C9orf72 ALS have failed in trials recently when treated patients performed worse than those on placebo. We hypothesize it may be safer to begin with the minimum dosing regimen reasonably hypothesized to confer benefit, rather than run the risk of dosing too frequently, and causing accumulated toxicity that masks clinical benefit.
3. **Time is of the essence.** Prion disease is without any therapeutic options, and patients are dying every day. In mid 2018, Ionis committed to developing an ASO for prion disease, and to date, a first in human dose has not yet been administered. Over a thousand patients in the U.S. and thousands more globally have died of prion disease in the four years we have waited, and we are contacted every week by people losing their loved ones and desperate to try anything that might help. Prion disease patients have nothing to lose, and we do not want to make the perfect the enemy of the good.

4.3.1. Study Protocol Outline

A proposed study protocol will be filed with IND submission. Our first draft study parameters are depicted in the table below. Our study is designed to select a patients whose symptomatic condition justifies the risk of this experimental new modality (measurable functional impairment with MRC Score[46] <20) but where survival time absent treatment is likely >3 weeks and thus sufficient to permit target engagement and potential for disease modification (MRC Score >13). We will select proposed low, medium, and high doses based on animal studies (Aim 4), with even the low dose being at a level predicted to measurably engage target and potentially modify disease. Based on our current understanding, we can estimate that these three dose levels may correspond to 50, 100, and 200 mg, but the actual levels will be chosen based on nonclinical data to be generated.

Patients will be randomized 2:1 drug to placebo. We will convene an Independent Data Monitoring Committee (IDMC) which will review the unblinded safety data after each dose level is completed and will make a determination as to whether to proceed to the next dose.. We will analyze 1- and 2-month post-dose CSF to determine target engagement and drug concentration, but these analyses will not be gating for dose escalation, which will be based on safety alone. The primary endpoint will be safety, secondary endpoints will be survival and CSF PrP lowering. Survival will be measured as time to either death *or* life-extending measures, since a substantial amount of variance in time to death in prion disease is due to differences in family decisions around end-of-life care[41,42]. CSF PrP lowering will be measured by ELISA as a percentage change relative to individual baseline to control for inter-individual differences in CSF PrP concentration[30].

Eligibility criteria (study population)	• Probable prion disease by CDC diagnostic criteria[43], including either:
	  o Positive spinal fluid RT-QuIC
	  o Positive *PRNP* mutation test
	• MRC Prion Disease Rating Score <20 and >13

Dose escalation schedule (study design)	• Low single dose - 2 patients • Placebo - 1 patient *IDMC determination based on interim safety data* • Mid single dose - 2 patients • Placebo - 1 patient *IDMC determination based on interim safety data* • High single dose - 2 patients • Placebo - 1 patient *IDMC determination based on interim safety data* • High* single dose - 4 patients • Placebo - 2 patients **Or highest dose deemed tolerated based on interim analyses.*
Additional interventions	CSF collection pre-dose and at 1 and 2 months post-dose for target engagement and pharmacokinetic analyses.
Primary endpoint	• Percentage of participants with adverse events, including procedure-related adverse events and adverse events occurring at or after initiation of treatment
Secondary endpoints	• Time to death or initiation of life-extending measures (permanent intubation or ventilation) • Time to drop 10 points on MRC Prion Disease Rating Scale • Cerebrospinal fluid PrP concentration
Duration of randomization	9 months
Data analysis method	Intention to treat

Table 1. Proposed clinical testing approach for first-in-human study. *This study design is not final; these draft high-level study parameters are provided to inform discussion of the choices made in the CMC and nonclinical sections below.*

4.4. Chemistry, Manufacturing and Control Data

a. Drug Substance

Divalent siRNA consists of covalently linked sense strands, each hybridized by Watson-Crick base pairing to an antisense strand; we refer to this as a triplex. The drug substance will be a lyophilized triplex in the form of a white powder. We are currently in the process of soliciting proposals from contract development and manufacturing organizations to produce the drug substance. The specifications that we have requested from these organizations are described below:

- *Analytical testing (sense strand intermediate)*

Attribute	Method	Acceptance Criteria
Identity	MS/MS	Sequence confirmed
Identity	RP-IP-UPLC	Retention time consistent with authentic material
Purity/impurities	RP-IP-UPLC	target 80% depending on process capability Report purity of sense strand Report total impurities Report impurities ≥ 0.2% as a function of RRT

- *Analytical testing (antisense strand intermediate)*

Attribute	Method	Acceptance Criteria
Identity	MS/MS	Sequence confirmed
Identity	RP-IP-UPLC	Retention time consistent with authentic material
Purity/impurities	RP-IP-UPLC	target 80% depending on process capability Report purity of antisense strand Report total impurities Report impurities ≥ 0.2% as a function of RRT

- *Analytical testing (release and retest)*

Attribute	Method	Acceptance Criteria
Appearance†	Visual inspection	White to pale yellow powder, free of visible foreign matter
Identity	SEC or non-denaturing IPRP-UPLC	Retention time consistent with authentic material
Identity	MS	Confirmed masses for sense and antisense sequences
Identity	Thermodynamic UV profile (Tm)	Report Result
Water content†	USP <921>	NMT 15%
Assay†	UV	Report
Purity/impurities of denatured triplex†	RP-IP-UPLC	Combined sense and antisense purity target 80% depending on process capability Report purity of each individual single strand Report total impurities Report impurities ≥ 0.2% as a function of RRT
Purity/impurities of triplex†	non-denaturing chromatographic method	~ 90% Report Total Impurities Report impurities ≥ 0.2% as a function of RRT
Sodium content (anhydrous)	ICP-OES	Target stoichiometric ~6.0%, NMT 10%
Bacterial Endotoxin†	USP <85>	Target <0.05 EU/mg
Bioburden†	USP <61>	TAMC ≤ 100 CFU/g TYMC ≤ 100 CFU/g

†stability indicating tests

Analytical Methods for drug substance

Demonstrate a UPLC method (RP-IP-UPLC) that provides suitable specificity for potential process impurities and degradation products (this method should be capable of using MS detection for characterization in early phase manufacture). This method should be capable of release and in-process testing.

Demonstrate a non-denaturing chromatographic method that provides suitable specificity for separating single strand intermediates from the drug substance. This method should be capable of release and in-process testing.

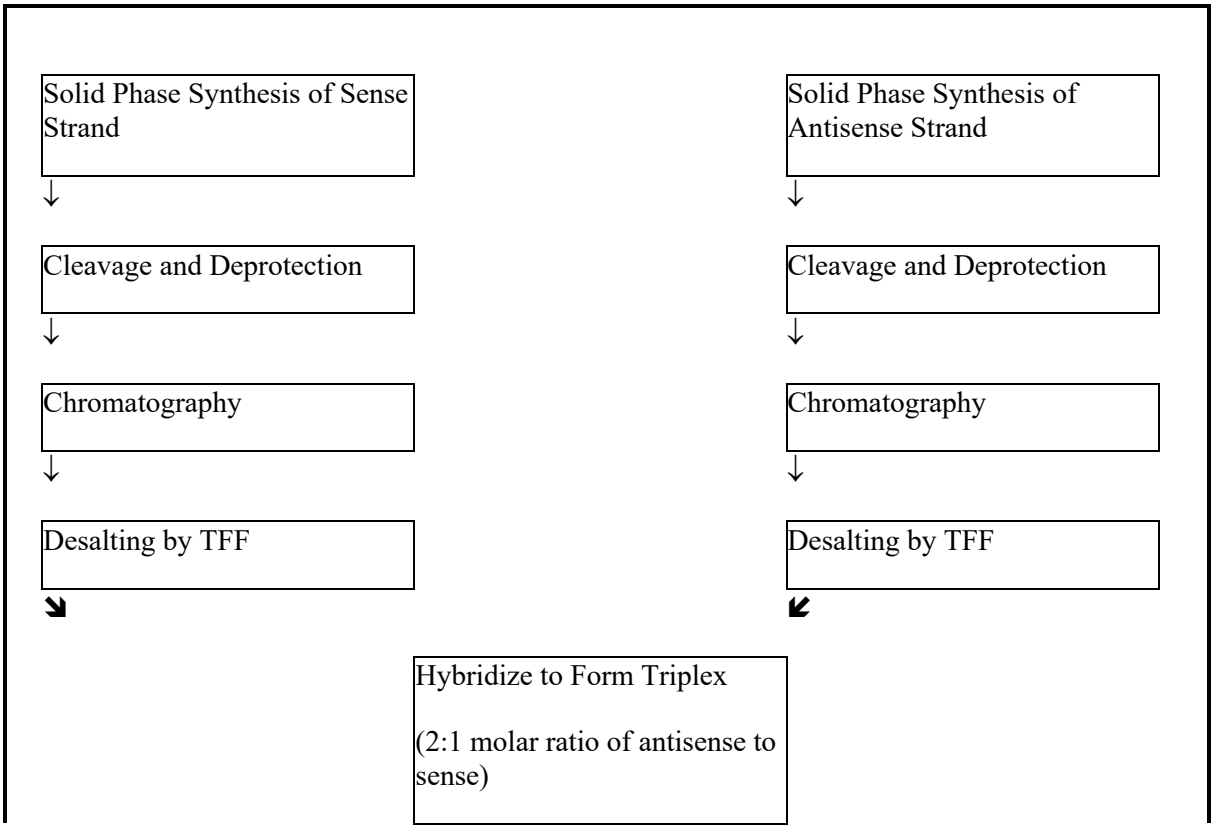
Supply of documented analytical procedures for both denaturing and non-denaturing methods.

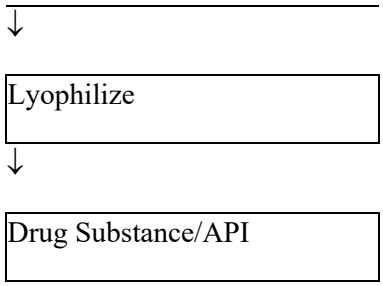
GMP manufacture

Manufacture drug substance, using client approved master batch records, according to cGMP using a scaled up process from Stage 1 targeting a yield of 15 g (net purity/water/sodium [label claim]). This is approximately 20g assuming 88% purity, 10% water and 6% sodium. If the values estimated for the quality attributes are considered to be different and may affect label claim then the gross manufacturing amount should be adjusted.

Define a process in consultation with Broad to supply drug substance for subsequent formulation into drug product for intrathecal administration to initiate a Phase 1 clinical trial.

Figure 1: Suggested Drug Substance Process Flow Diagram





Precautions should be taken to prevent the microbial contamination of the process streams to minimize the endotoxin and bioburden levels in the drug substance. Measures should include using a sterile, depyrogenated or sanitized product contact path. It is understood that this is for the manufacture of a non-sterile drug substance but endotoxin levels must be minimized as much as possible.

Pool, UF, hybridize and ionic condition material in conjunction with sponsor at key decision points to provide material for toxicology and formulation into drug product.

Delivery of executed batch records that document the manufacture of the drug substance. The records should include information describing quality attributes (CoA) of the starting materials (contribute to the structure of the drug substance), details of the manufacture, any deviations from Standard Operating Procedures (including the investigations and the impact of the deviation on the quality of the product), analytical raw data supporting the CoA of the drug substance, BSE/TSE statement and GMP (phase appropriate) declaration.

Stability

Drug substance (see stability indicating tests) will be laid down at the storage temperature to enable performance of two retests for assigning retest dates.

Stability data when generated should be provided to the client in a tabulated form or a "retest CoA".

b. Drug Product

We envision that the Drug Product will ultimately consist of Drug Substance triplex dissolved in sterile saline with divalent cations Ca^{2+} and Mg^{2+}.

The Drug Substance will undergo ionic conditioning to saturate divalent cation binding sites, which improves tolerability upon injection into CSF[44]. Drug Substance will be mixed with a molar ratio of 14.3 units of Ca^{2+} to 1.7 units of Mg^{2+} to 1 monovalent equivalent of siRNA (0.5 units of divalent siRNA). It will then be diluted in saline and stored in sterile vials.

If the maximum dose to be tested in humans is determined in nonclinical studies to be 200 mg, then one vial for clinical use might contain 200 mg of drug in 10 mL of saline. Our draft clinical plan calls for 10 patients receiving active drug, corresponding to 10 vials, and our nonclinical studies described below could be completed with an additional 10 vials. We calculated that a standard testing battery of release, stability, and retains would require 170 vials of Drug Product. Thus, if all vials are required to be of an identical size, testing would require nearly 10 times as much Drug Product as will be

administered in humans and animals, in turn increasing the amount of Drug Substance required by approximately 10-fold. In order to make this project feasible for us to undertake without a commercial sponsor, we propose that the Drug Product could be prepared in differently sized vials, where vials for clinical use could contain 200 mg in 10 mL, while vials intended for testing could contain as little as 10 mg in 500 µL.

Anticipated Process Changes for Manufacture of Investigational Product

No process changes are anticipated. We plan to produce one batch that will suffice for our planned nonclinical and clinical studies.

4.5. Pharmacology and Toxicology Information

All of the nonclinical studies performed to date are described above. Below we list the additional nonclinical studies we propose to perform prior to IND filing. We are proposing to perform pharmacology and toxicology studies in rodents alone, with GLP toxicology studies only in rats, in which the compound is not sequence matched. We considered the possibility of performing nonclinical studies in cynomolgus macaques, in whom the compound is sequence matched. This would have the advantage of enabling assessment of biodistribution and target engagement in a large brain after intrathecal dosing, but considerable disadvantages in terms of delay, feasibility, and study complexity. We proposed a small set of both non-GLP and GLP macaque studies to NIH, but we do not yet know whether the NIH grant is awarded nor whether NIH will be able to support such studies through its URGenT contractor network; even in the best case, these studies could prove rate-limiting for initiating human studies. On balance, given the rapid lethality of prion disease and the lack of any alternative treatment option, we propose that initiating human dosing based on rodent toxicology alone would be justified, and indeed, is the option that best honors our commitment to the patient community.

Non-GLP off-target analysis in human cell lines. Will seek to characterize potential hybridization-dependent off-target effects of our lead in the manner that would be suggested by guidance if we were developing an N-of-1 ASO[45]. In order to empirically identify off-target effects of our lead, we will test its gene silencing profile by RNA-seq in a panel of brain-derived human cell lines. Compound at 0.5 µM, or no compound control, will be transfected in biological quadruplicate into IMR32 and SH-SY5Y (neuronal) and U-251 MG, U-87 MG, and T98G (glial) cell lines. Cell pellets will be submitted for whole transcriptome sequencing to identify differentially expressed genes.

Non-GLP pharmacology modeling. We will conduct non-GLP PD/PK studies in humanized Tg26372 mice to establish the relationship between tissue drug concentration and target engagement in order to facilitate dose selection at later project stages. Groups of N=8 Tg26372 mice (mixed sex) will receive ICV doses of 0, 0.1, 0.3, 1, 3, and 10 nmol 2439-exNA and be sacrificed 30 days post-dose (total: 48 animals). 6 brain regions (prefrontal cortex, striatum, thalamus, hippocampus, visual cortex, and cerebellum) will be dissected (total: 288 samples). In each sample we will quantify *PRNP* mRNA by qPCR (normalized to *Tbp* and further normalized to the mean of 0-dose animals) and drug concentration using the in-house version of our PK assay (Aim 2.2). We will fit four-point hill slope dose response curves for each region to correlate drug concentration (µg/g tissue) with knockdown (percent).

GLP pharmacology and toxicology. We propose to conduct GLP pharmtox studies in rats only. All dosing will be intrathecal (IT) and all animals will undergo full necropsy for a standard battery of safety and toxicology endpoints and brain tissue drug concentration determination.

Species	Route	N (M/F)	Dose level	Dosing timepoints	Harvest timepoint
Rat	IT	4/4	0 nmol	0 weeks	4 weeks
Rat	IT	4/4	0.3 nmol	0 weeks	4 weeks
Rat	IT	4/4	1.0 nmol	0 weeks	4 weeks
Rat	IT	4/4	3.0 nmol	0 weeks	4 weeks
Rat	IT	4/4	10.0 nmol	0 weeks	4 weeks
Rat	IT	4/4	30.0 nmol	0 weeks	4 weeks
Rat	IT	4/4	100.0 nmol	0 weeks	4 weeks
Rat	IT	4/4	0 nmol	0 and 13 weeks	17 weeks
Rat	IT	4/4	0.3 nmol	0 and 13 weeks	17 weeks
Rat	IT	4/4	1.0 nmol	0 and 13 weeks	17 weeks
Rat	IT	4/4	3.0 nmol	0 and 13 weeks	17 weeks
Rat	IT	4/4	10.0 nmol	0 and 13 weeks	17 weeks
Rat	IT	4/4	30.0 nmol	0 and 13 weeks	17 weeks
Rat	IT	4/4	100.0 nmol	0 and 13 weeks	17 weeks

Table 2. Proposed GLP nonclinical studies. *IT = intrathecal. Note that dose levels shown are tentative, to be adjusted pending results of PD/PK modeling in Aim 3.2 and non-GLP pilot in Aim 3.3.*

4.6. Summary of Previous Human Studies

None.

5. Specific Objectives/Outcomes Expected from the Meeting

Concurrence on high-level clinical plan focused on symptomatic patients.

Concurrence on CMC and testing requirements for this clinical plan.

Concurrence on nonclinical studies required for this clinical plan.

6. Proposed Agenda

- Introduce participants (5 minutes)
- Sonia Vallabh: brief introduction to this project and our patient community (5 minutes)
- Discussion of questions to the agency and any other input from agency (40 minutes)
- Summary and action items (10 minutes)

7. Questions

This section provides the questions posed by SPONSOR to the FDA. For ease of review, the questions are provided in **bold font,** and SPONSOR'S position regarding each question is provided in normal font below the question.

1. [Clinical] We propose to test a single dose in symptomatic patients, escalating dose after an independent committee evaluates safety data, but not waiting for biofluid analysis results to become available.

   Is the proposed dose escalation schedule and monitoring program adequate?

2. [Preclinical] We propose to conduct GLP toxicology and pharmacokinetic studies only in rats, supplemented by non-GLP pharmacokinetic/pharmacodynamic studies in mice.

   Does the agency agree that data from these studies would be appropriate to support the proposed clinical plan?

3. [CMC] We propose to generate Drug Product vials of different sizes, with smaller aliquots used for testing than dosing, to maximize utilization of manufactured drug.

   Does the agency agree that this plan is acceptable?

8. Sponsor participants

Sonia Vallabh, PhD — Broad Institute
Eric Minikel, PhD — Broad Institute
Eric Lander, PhD — Broad Institute
Steven E. Arnold, MD — Massachusetts General Hospital
Brian S. Appleby, MD — National Prion Disease Pathology Surveillance Center

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